Chapter- 5

~~Marginal productivity theory of labor demand~~

The theory of labor demand in the long run

→ Capital and labor are variable.

Technology of production: Isoquants

Goal of firms: Profit maximizing output production
↓
least cost production method and
appropriate combination of K and L.

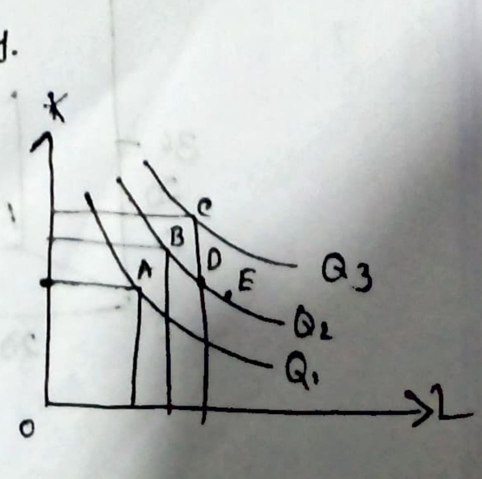
Influenced by:
→ technological constraints places on the mix of K and L.
→ relative prices of the factor inputs.

Firm's production function, $Q = f(K, L)$.

→ Expresses the relationship between the level of K and L inputs and the maximum obtainable level of output, given the current state of technology.

Different combinations of K and L for alternative production process:-

Isoquants:-



→ ISoquant: all the various combinations of K and L, that are just able to produce a given level of output.

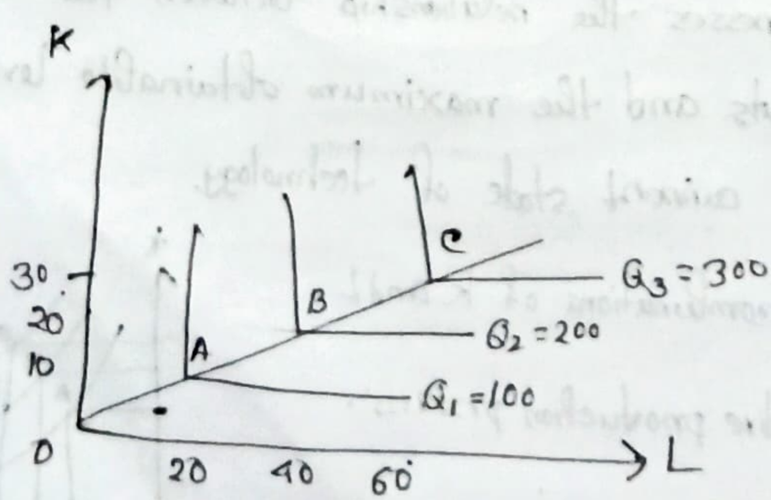
Features: (1) Convex to the origin.

In moving from point B to D in Q_2 : labor must be increased and capital must be decreased.

Giving rise to a diminishing MRTS (the additional amount of K that is required if labor is changed by 1 unit and Q is kept constant).

MRTS is given by the negative slope of the isoquants.

If the technology is such that K and L must be used in strict proportion to each others, allowing one and only K/L ratio in production. Example: Air lines.



For L shaped isoquants:- (1) Zero substitution possible between K and L, MRTS = 0.

(2) If one factor is fixed, adding more of the other doesn't increase Q.

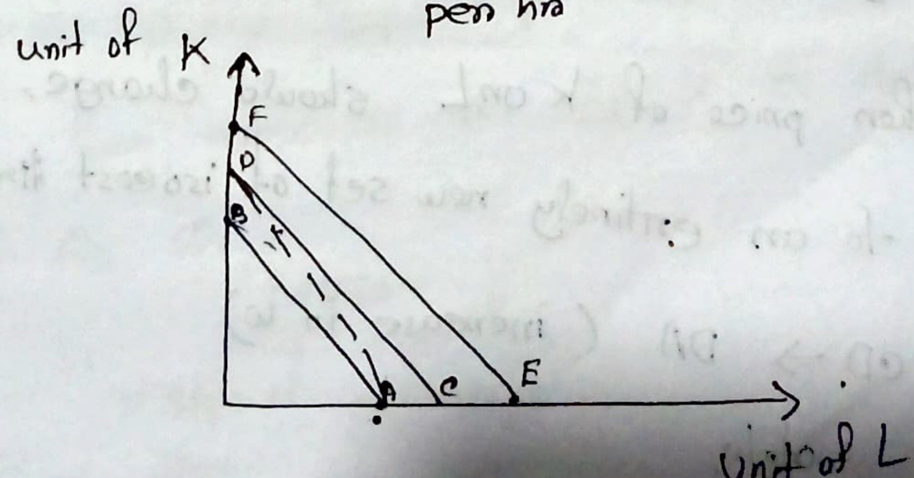
(3) The ray from the origin through the kink points of isoquants, the slope of the ray $= \frac{K}{L}$.

Factor prices:- Isocost lines

To choose the particular combination to maximize output the relative prices of K and L must be known.

Rental rate or price for a unit of labor $= R_1$

wage rate of labor per hr $= W_1$



Isocost line $\rightarrow$ various combinations of K and L that the firm could purchase with a particular amount of money expentiture (E).

$$\frac{E_1}{W_1} = L_1 \quad \text{and} \quad \frac{E_1}{R_1} = K_1$$

(Point-A) (Point-B)

Features:-

① Slope of the iso-cost line is constant and equal to the negative ratio of factor prices $= -\frac{W_1}{R_1}$

② For given factor prices, a higher level of expentiture by the firm on K and L will give rise to an entirely new isocost line, that is parallel and to the right of the original one. AB → CD

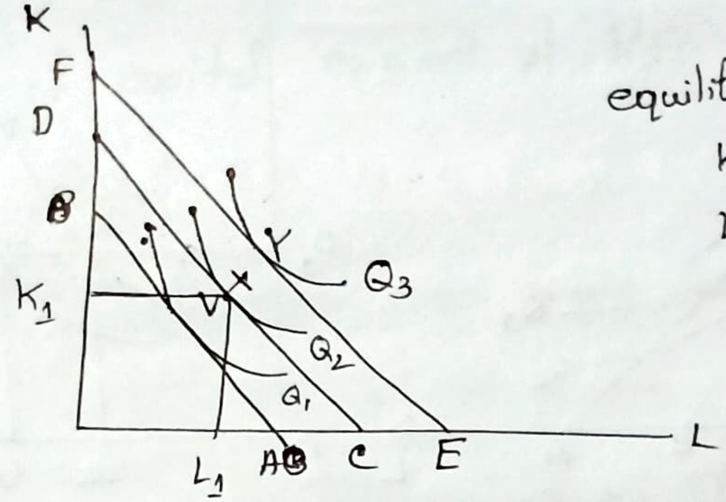
③ If either price of K or L should change, it will give rise to an entirely new set of isocost lines.

If, CD → DA (increase in W)

rotate.

Q: Equilibrium level of employment:-

Combination of isoquant and isocost lines:-



equilibrium: at point X

K_1 amount of capital

L_1 amount of labour

For profit maximization, $P = MC$.

Suppose, at Q_2, $P = MC$.

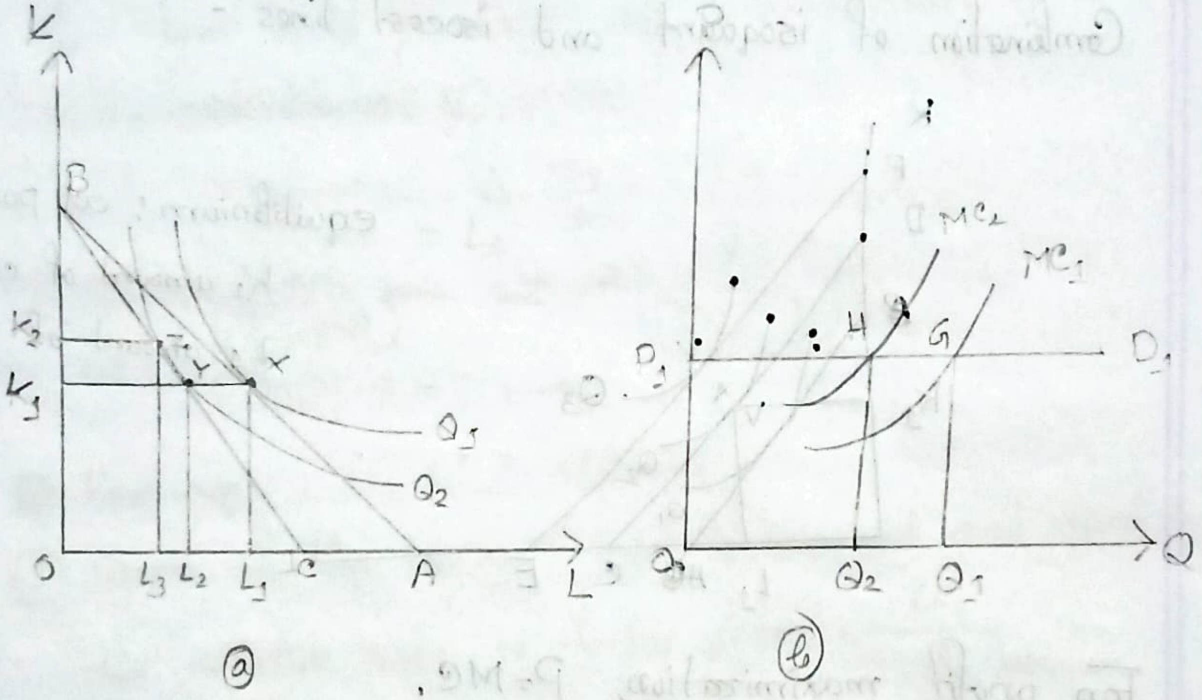
How to produce Q_2 →

what combination of K and L will be cost minimizing →

Isocost CD and Isoquant Q_2, is tangent at point X, where the firm minimizes cost. At X, the slope of CD = slope of Q_2. so,

$$\underset{\text{Slope of Isoquant}}{MRTS} = \underset{\text{Slope of isocost}}{\frac{w}{R}}$$

A change in the wage rate:-



At initial level:-

wage rate – W_1

cost of capital – R_1

so, isocost line - AB

tangency between isocost AB and isoquant Q_1 is at point X,

at point X → least cost exployment - L_1 with capital K_1

in ⓑ, Demand curve for perfectly competitive market D_1 with MC_1, tangent to point G the firm will produce at point G as $P_1 = MC_1$.

Increase in wage rate from W_1 to W_2:-

Cost of capital constant at R_1.

Two effects:-

① intial impact:- increase in w rotate the isocost line left from AB to CB in ⓐ.

As the cost of labor is higher, this will increase the firm's cost of production, shifting the MC curve to the left from MC_1 to ~~M2~~ MC_2, lowering the output from Q_1 to Q_2, where, $P_1 = MC_2$.

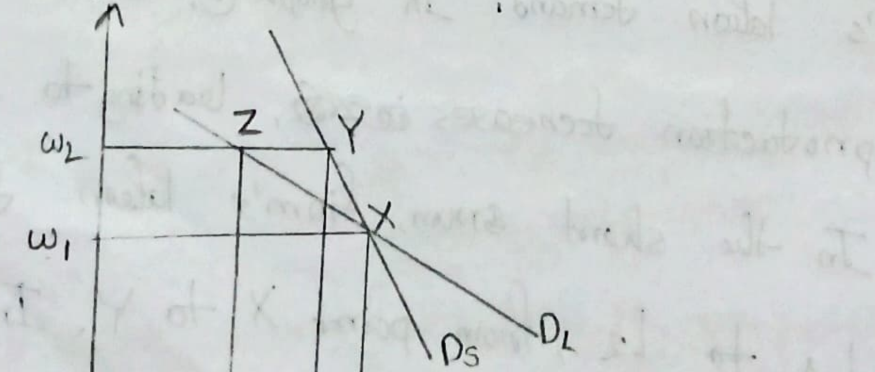
the Reduction in the level of Q leads to a decrease in the firm's labor demand. In graph ⓐ, after increasing the w, production decreases ~~...~~, leading to a new isoquant Q_2. In the short run, firm's labor demand falls from L_1 to L_2, from point X to Y. Input combination is K_1, L_2 with output Q_2. This is scale effect.

② Long run impact:- Given the higher wage w_2, the cost-minimization condition is,

$$MRTS = \frac{w_2}{R_1}.$$

This occurs at point Z, at the tangency of BC and Q_2. In the short run, the firm is constrained to use K_1 and thus cannot reach to Z. But, in the long run, the higher price of labor will influence the firms to substitute capital for labor for producing Q_2, increasing capital to K_2 and decreasing labor to L_3. This is substitution effect.

Q:- Long run and short run demand curves for labor:-



The long run labor demand curve D_L is more elastic than short run labor demand curve. In the long run, the ability to change not only the level of production but also the amount of K provides the firm with much more flexibility in adapting to higher or lower wage, resulting in a demand curve for labor that is more sensitive to changes in wage costs in the long run than in the short run. In the graph, a rise in the wage from W_1 to W_2 leads to a decline in labor demand in the short run from L_1 to L_2 (Point X to Y) due to the scale effect. In the long run, labor demand declines further from L_2 to L_3 (Point Y to Z) due to substitution effect.

Q: Determinants of the elasticity of labor demand:-

The four laws of derived demand states that, the demand for labor, other things equal, will be more elastic:-

1. The larger the price elasticity of demand for the product being produced.

2. The greater the share of labor cost as a percentage of the total cost of production.

3. The greater the ease of substitution in production between L and other factors inputs such as, K.

4. The greater the elasticity of supply of other competing factors of production.

Technological change and labor demand:-

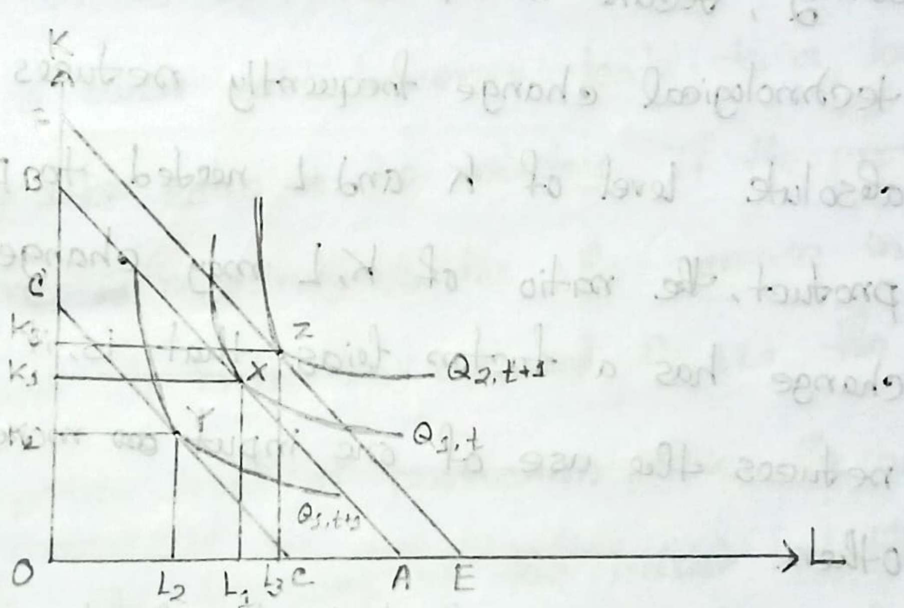
Technological change has two effects:-

1. The initial effect is to reduce the demand for labor as better technology allows firms to produce a given level of output with fewer workers.

2. The improved technology results in lower production costs and, thus, lower product prices, increased sales and a greater demand for labor.

Q:- The displacement of labor by technological change:-

Technological change → new, more efficient ways to produce a product with less labor or capital.



Technological change shifts the isoquant $Q_{1,t}$ to $Q_{1,t+1}$, these two represents exactly the same level of outputs; the difference is that with the advance in the state of technology, Q_1 can now be produced in period $t+1$ with less L and K.

If factors prices remained at W_1, R_1, firms in this industry would minimize the costs of producing Q_1 by using only K_2 units of capital and L_2 units of labor given by tangency of the isoquant $Q_{1,t+1}$ and the isocost line CD (Point-Y)

In this case, the demand for labor is reduced from L_1 to L_2 (Point X to Y), it also apears to be capital saving, decline in capital from K_1 to K_2. While technological change frequently reduces both the absolute level of K and L needed to produce a particular product, the ratio of K, L may change if technological change has a factor bias, that is, it it proportionately reduces the use of one input more than the other.

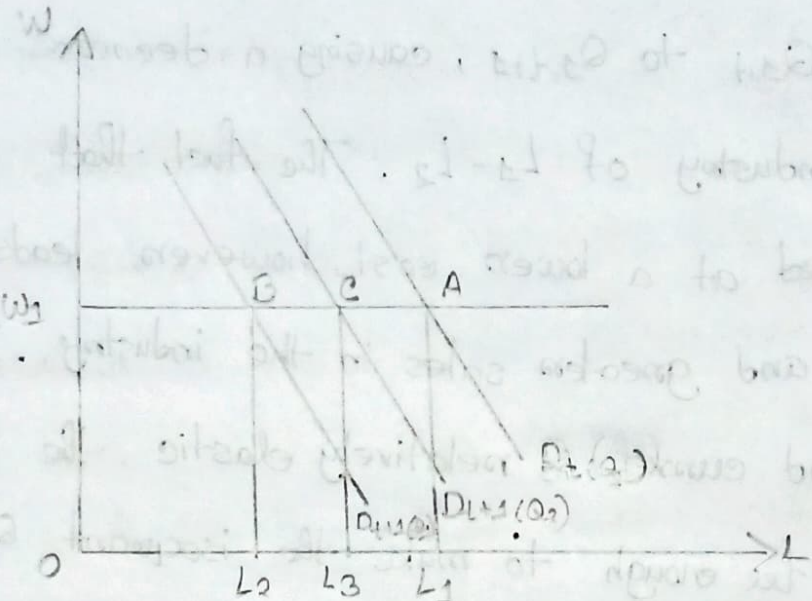
If the only effect of technological change was to continually shift the isoquants toward the origin, over time the economy would need fewer and fewer workers to produce a given level of output.

Q:- Technological change and product demand:-

Technological change leads to an increase both in the sales of the individual industry and in the total product demand in the entire economy.

The initial impact of technological change shifts the isoquants from $Q_{1,t}$ to $Q_{1,t+1}$, causing a decrease in employment in the industry of $L_1 - L_2$. The fact that Q_1 can now be produced at a lower cost, however, leads to a lower price and greater sales in the industry. If the product demand curve is relatively elastic, the increase in sales may be enough to make the isoquant $Q_{2,t+1}$, the new profit maximizing level of production. Given the same ratio of factor prices, the tangency of the isocost line EF and isoquant $Q_{2,t+1}$ (Point Z) yields a new demand for labor L_3 and demand for capital K_3. While $L_1 - L_2$ workers initially lost their jobs due to the more efficient technology, the lower price this made possible resulted in the end in a net increase of employment in the industry of $L_3 - L_1$.

The combined effect:-



At the wage rate of W_1, the long run labor demand curve $D_t(Q_1)$, the output is Q_1 and the demand for labor is L_1 (Point-A). The first impact of technological change is to displace labor as better technology enables the firm to produce the same Q_1 units of output with less labor. This shifts the Long run demand curve left from $D_{t(Q_1)}$ to $D_{t+1}(Q_1)$ (Point-B). The resulting decline in the production price, however, stimulates an increase in sales to Q_2, leading to a righthand shift of the labor demand curve to whether this subsequent increase in labor demand of $L_3 - L_2$ is enough to offset the initial displacement of labor of $L_1 - L_2$, depends on the extent to which the savings in PC are passed on to buyers in the form of lower prices and on the price, income elasticity of the industry's product demand curve.

$D_{t+1}(Q_2)$ (Point:- e). Here, sales do not increase enough to offset the original displacement of labor and employment suffers a net decline of $L_1 - L_3$. The opposite could also occur.

Long run equilibrium level of employment:-

Given:- $Q = f(K, L)$

Labor demand, $L = l(R, W, Q)$

(a) Show that long run equilibrium of employment is determined where $MRTS = \frac{W}{R}$ / Condition for eq. level of L.

(b) ~~Derive~~ Show that, the slope of isoquant is MRTS.

(c) Show that, the long run effect on labor demand for a wage change decomposing substitution and scale effect.

Ans:-

ⓐ The profit function,

K = capital
R = Price of capital
L = Labor
w = wage rate.

$$\pi = P.Q - KR - \omega L$$

$$= P.f(K,L) - KR - WL$$

The FOC requires,

$$P.\frac{\partial f}{\partial k} - R = 0$$

$$\Rightarrow P = \frac{R}{\partial f/\partial K} \quad \text{———— (i)}$$

$$P.\frac{\partial f}{\partial L} - w = 0$$

$$\Rightarrow P = \frac{\omega}{\partial f/\partial L} \quad \text{———— (ii)}$$

So, we can write from (i) and (ii),

$$\frac{\omega}{R} = \frac{\partial f/\partial L}{\partial f/\partial k}$$

$$\Rightarrow \frac{\omega}{R} = MRTS$$

$$\Rightarrow MRTS = \frac{\omega}{R} \quad \text{———— (iii)}$$

Here, the ratio $\frac{\partial f/\partial L}{\partial f/\partial k}$ indicates how much additional output a firm can produce ~~for~~ from ~~one~~ one more unit of Labor (MP_L) relative to the additional output lost by giving up one more unit of capital (MP_K).

The eq③, satisfies the condition of profit maximization given the product price and input prices. This is the same condition for equilibrium level of employment.

ⓑ given that,

$$Q = f(K, L)$$

totally differentiating,

$$df = \frac{\partial f}{\partial k} \cdot dk + \frac{\partial f}{\partial L} dL \quad \text{------} \quad (i)$$

Since, output does not change as one move along an isoquant, the (i) can be re-written as,

$$0 = \frac{\partial f}{\partial k} dk + \frac{\partial f}{\partial L} \cdot dL$$

$$\Rightarrow \frac{\partial f}{\partial L} \cdot dL = -\frac{\partial f}{\partial k} dk$$

$$\Rightarrow \frac{\frac{\partial f}{\partial L}}{\frac{\partial f}{\partial k}} = -\frac{dk}{dL}$$

$\Rightarrow$ MRTS = –(Slope of the isoquant)

So, $-\boxed{\text{Slope of the isoquant}} = \frac{w}{R} = MRTS$

(c) The labor demand,

$$L = l(R, w, Q)$$

$$\text{or, } L = l\left[R, w, ~~\ldots~~ Q(P(R,w))\right] \quad \text{--- (i)}$$

Totally
Differentiating (i),

$$dL = \frac{\partial l}{\partial R} dR + \frac{\partial l}{\partial w} \cdot dw + \frac{\partial l}{\partial Q} \cdot \frac{\partial Q}{\partial P} \cdot \frac{\partial P}{\partial R} \cdot dR +$$

$$\frac{\partial l}{\partial Q} \cdot \frac{\partial Q}{\partial P} \cdot \frac{\partial P}{\partial w} \cdot dw$$

Holding cost of capital, R constant,

$$dL = \frac{\partial l}{\partial w} \cdot dw + \frac{\partial l}{\partial Q} \cdot \frac{\partial Q}{\partial P} \cdot \frac{\partial P}{\partial w} \cdot dw$$

$$\Rightarrow dL = dw\left(\frac{\partial l}{\partial w} + \frac{\partial l}{\partial Q} \cdot \frac{\partial Q}{\partial P} \cdot \frac{\partial P}{\partial w} \right)$$

$$\Rightarrow \frac{dL}{dw} = \underbrace{\frac{\partial l}{\partial w}}_{\text{Substitution effect}} + \underbrace{\frac{\partial l}{\partial Q} \cdot \frac{\partial Q}{\partial P} \cdot \frac{\partial P}{\partial w}}_{\text{scale effect}}$$

So, total change in employment due to wage change have two effects.